

Nisarg Vaghela

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EDUCATION

National Institute of Technology, Surat

2024 – Present

B.Tech in Artificial Intelligence; CGPA: 7.00 (Current)

Surat, Gujarat

- **Relevant Coursework:** Data Structures, Design and Analysis of Algorithms, DBMS, Computer Organization, Discrete Mathematics, Linear Algebra.

PROJECTS

ChatPDF – AI-Powered Document Assistant | *FastAPI, Next.js, LlamaIndex, FAISS, MongoDB, Groq*

[GitHub](#)

- Developed a production-ready RAG application using **FastAPI**, **Next.js 14**, **LlamaIndex**, and **FAISS** for vector similarity search on PDFs.
- Built a split-view UI with an integrated PDF viewer and intelligent, clickable citations routing to specific source pages.
- Engineered secure per-user data isolation using MongoDB for chat histories and Supabase for storage, secured via JWT.
- Integrated **Groq** (LLaMA 3.3 70B) for optimized document parsing and sub-second context generation.

Basho by Shivangi – E-commerce Pottery Platform | *Next.js, Firebase, Razorpay, Tailwind*

[GitHub](#)

- Developed a scalable D2C e-commerce platform using **Next.js 14** and **Firebase** with product catalogs and a seamless checkout flow.
- Integrated **Razorpay** for secure payments and the **Resend** API for automated order confirmation emails.
- Engineered an admin panel with **Recharts** for real-time inventory management, workshop bookings, and sales analytics.
- Designed a responsive UI using **Tailwind CSS** and **Framer Motion**, achieving a 91% Lighthouse performance score.

Testimonial Studio – AI Video Platform | *Next.js, Gemini API, Deepgram, FFmpeg, Cloudinary*

[GitHub](#)

- Engineered an interactive AI host using the **Gemini API** for dynamic questioning and **Deepgram** for real-time speech transcription.
- Architected a video pipeline utilizing Groq LLaMA 3 to analyze transcripts and **FFmpeg** to automatically generate social media-ready reels.
- Developed a custom browser-based recording engine using the MediaRecorder API to simultaneously capture user video and a synchronized AI avatar.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript, HTML5/CSS3

Frameworks & Libraries: Next.js, React, FastAPI, Node.js, LlamaIndex, FAISS, NumPy, FFmpeg

DevOps, Tools & Databases: Git, MongoDB, Firebase, Supabase, Vercel, Cloudinary, Deepgram, Groq

ACHIEVEMENTS

Hackathon Winner: Secured **3rd Prize** for building **PowerMind**, an AI-powered financial data assistant utilizing LlamaIndex, FAISS, and Groq for intelligent document parsing and querying.

Hackathon Participations: Actively competed and built projects at [Daiict - Hackout](#), [Odoo X GCET](#), and [Hack The Tank - SVNIT](#).

Competitive Programming: Active participant on [Codeforces](#) and [LeetCode](#) (Contest Rating: **1374**).

Problem Solving: Solved **algorithmic problems** across various coding platforms, demonstrating strong grasp of Data Structures and Algorithms.

POSITIONS OF RESPONSIBILITY

Google Developer Group (GDG), NIT Surat | *Junior Developer (Technical Expertise)*

Oct 2025 – Present

Association for Computing Machinery (ACM), NIT Surat | *Executive (Organizing Tech Events)*

Sept 2025 – Present

Cutting Edge Visionaries (CEV), NIT Surat | *Executive (Organizing Tech Bootcamps)*

Aug 2025 – Present